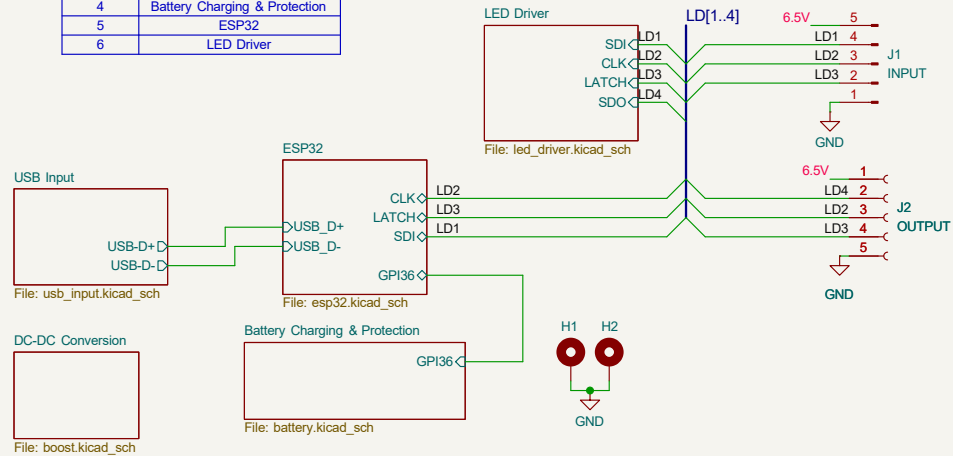


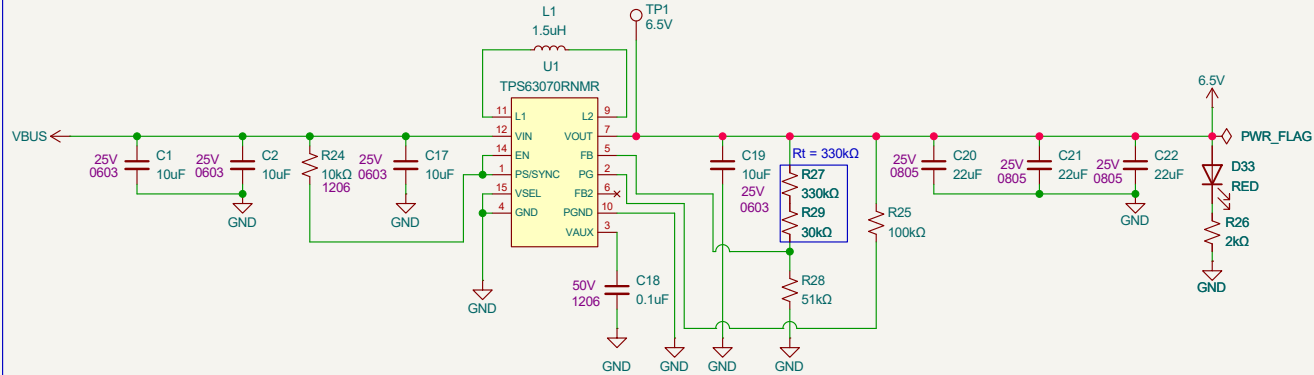
| Page # | Content                       |
|--------|-------------------------------|
| 1      | Root                          |
| 2      | DC-DC Conversion              |
| 3      | USB Input                     |
| 4      | Battery Charging & Protection |
| 5      | ESP32                         |
| 6      | LED Driver                    |



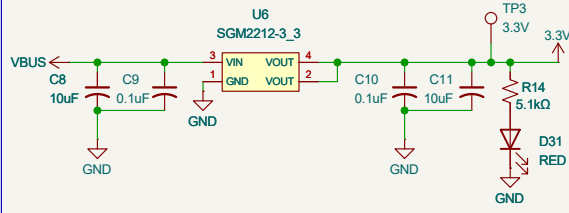



|                     |                            |           |
|---------------------|----------------------------|-----------|
| File: v2.kicad_sch  | Confidential & Proprietary |           |
| Size: User          | Date: 2026-09-25           | Rev: v2.0 |
| KiCad E.D.A. 10.0.2 | Page: 1/6                  |           |

VBAT -> 6.5V

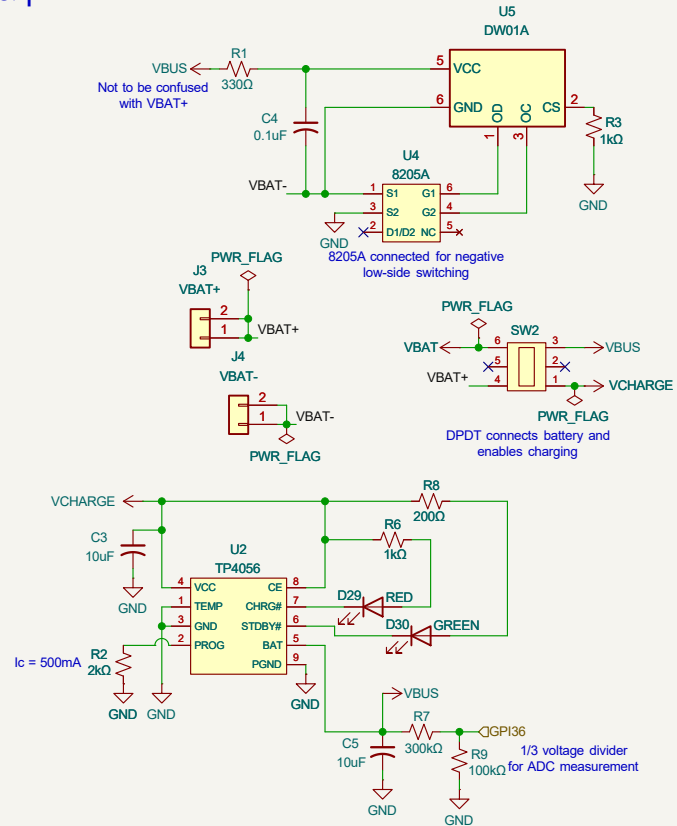


VBAT -> 3.3V



[illegible]

## Battery charging & protection



**BIRFLM**

File: battery.kicad\_sch Confidential & Proprietary

Size: User

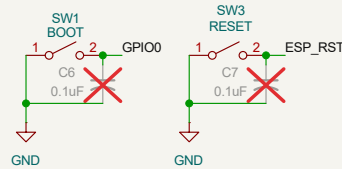
Date: 2026-09-25

Rev: v2.0

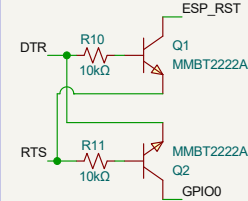
KiCad E.D.A. 10.0.2

Page: 4/6

## Boot & Reset

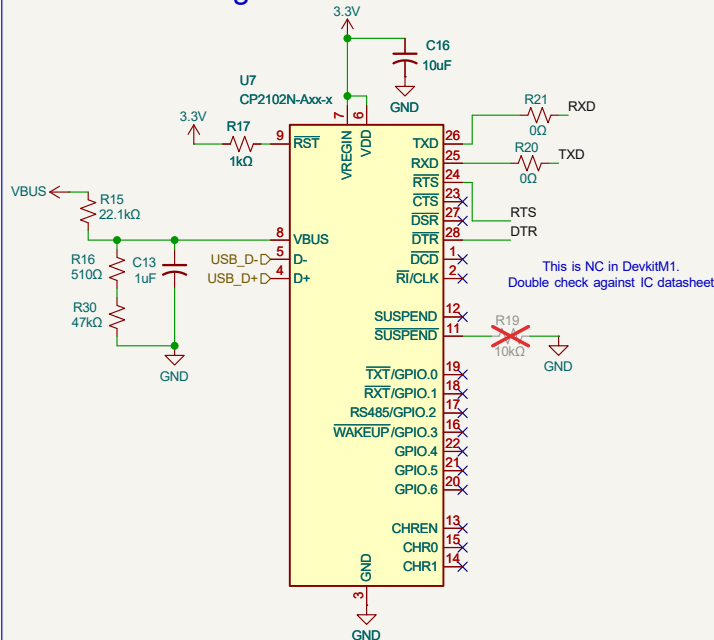


## Auto-Reset Circuit

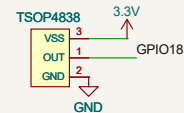
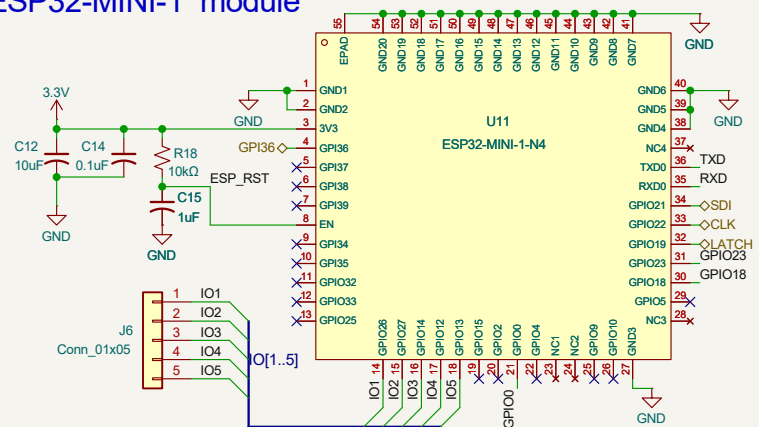


| DTR | RTS | ESP_RST | IO0 |
|-----|-----|---------|-----|
| 1   | 1   | 1       | 1   |
| 0   | 0   | 1       | 1   |
| 1   | 0   | 0       | 1   |
| 0   | 1   | 1       | 0   |

## USB-UART Bridge



## ESP32-MINI-1 module



File: esp32.kicad\_sch Confidential & Proprietary

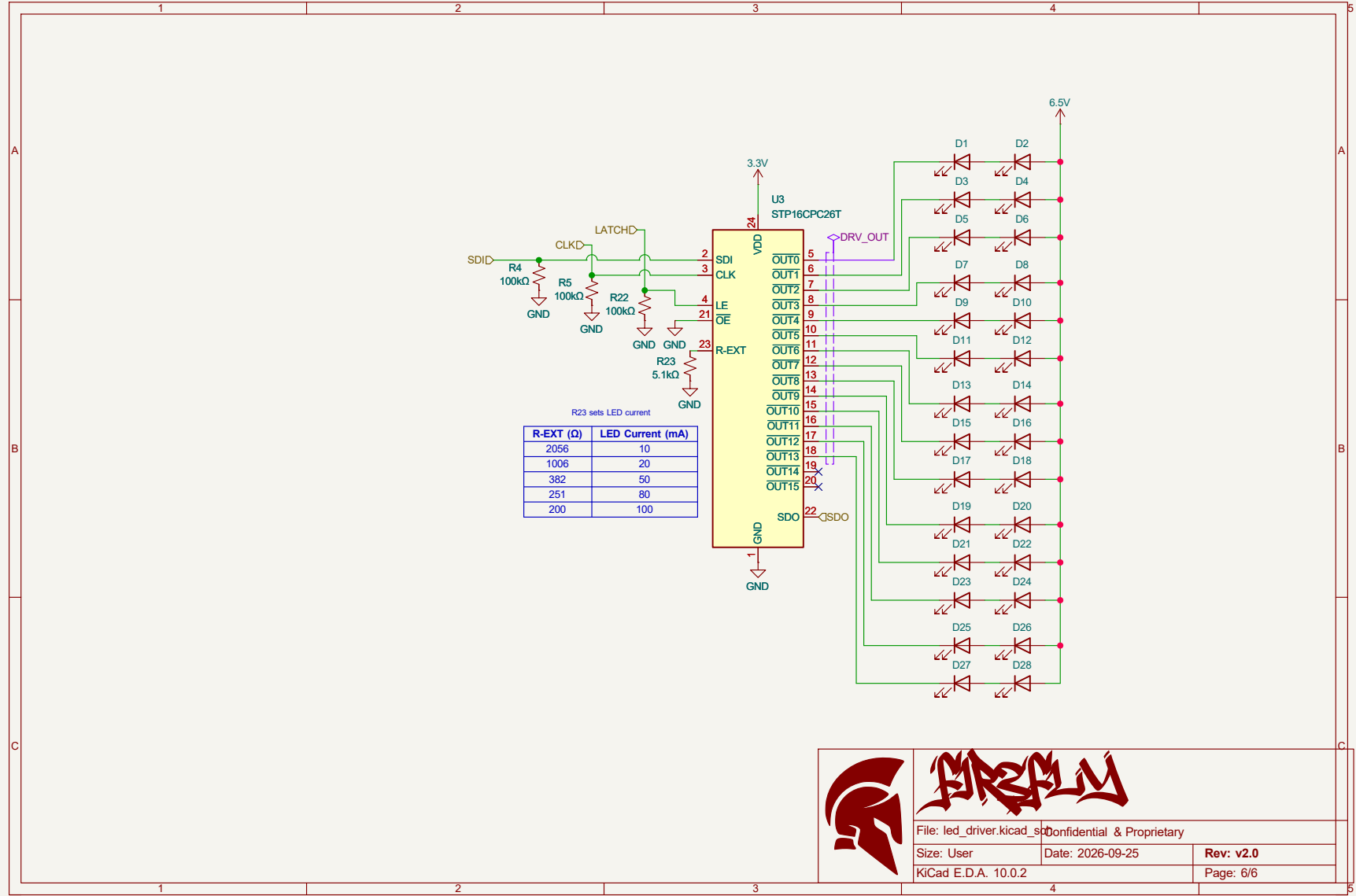
Size: User

Date: 2026-09-25

Rev: v2.0

KiCad E.D.A. 10.0.2

Page: 5/6



**BIRFLM**

File: led\_driver.kicad\_sch Confidential & Proprietary

Size: User

Date: 2026-09-25

Rev: v2.0

KiCad E.D.A. 10.0.2

Page: 6/6